

PREDICTING NEXT-QUARTER DEBT SERVICING RISK FOR PHILIPPINE COMPANIES

CORPORATE SOLVENCY AND PREDICTIVE MODELING

Predicting Next-Quarter Debt Servicing Risk for Philippine Companies

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THE BUSINESS CHALLENGE



Debt servicing stress emerges before it's visible.
Financial distress can escalate quickly and impact lenders, investors, and the economy.



Traditional monitoring is reactive.
Relies on historical analysis and manual review—slow and resource-intensive.



Need for predictive, data-driven insights.
Identify high-risk companies early to act before problems become real defaults.



Better risk decisions. Better business outcomes.
Protect capital, improve portfolios, and support financial stability.

OUR AI SOLUTION



Integrated Multi-Source Data
Financial statements, market data, company disclosures, and macroeconomic indicators.



Advanced Machine Learning
Random Forest (best model) analyzes patterns across 29 key financial and economic drivers.



Explainable & Transparent
SHAP explainability shows what drives each prediction—building trust and confidence.



Early Warning in Action
Predicts next-quarter risk to help you act early, not react late.

CORPORATE DEBT SERVICING CHALLENGES

THE BUSINESS PROBLEM

- Listed non-financial firms face critical debt servicing risks requiring timely identification.
- Traditional ratios assume linear relationships, failing to capture complex risk patterns.
- Manual review processes are too time-consuming for proactive risk mitigation.

NECESSITY OF AI-DRIVEN SYSTEMS

- AI detects subtle, non-linear interactions among diverse financial indicators.
- Provides an early warning system to overcome limitations of manual oversight.

MACROECONOMIC CONTEXT

- Rising interest rates and inflation pressure corporate repayment capacities.
- Exchange rate fluctuations (USD/PHP) significantly impact debt servicing costs.

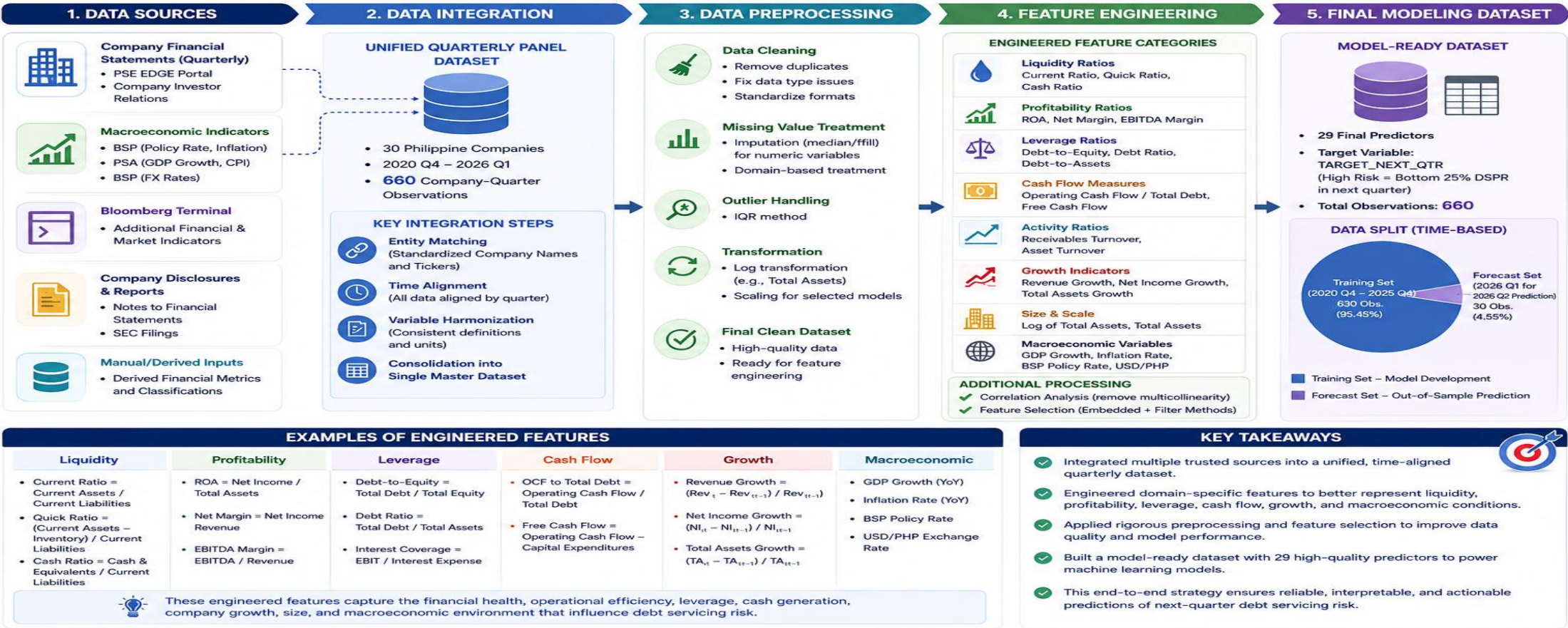
STUDY PARAMETERS

Bottom 25%
DSPR Threshold for "High Risk" Classification

2020 Q4 – 2026 Q1
Quarterly Time Horizon for Predictive Analysis

DATA INTEGRATION & FEATURE ENGINEERING STRATEGY

Building Predictive Features that Drive Next-Quarter Debt Servicing Risk



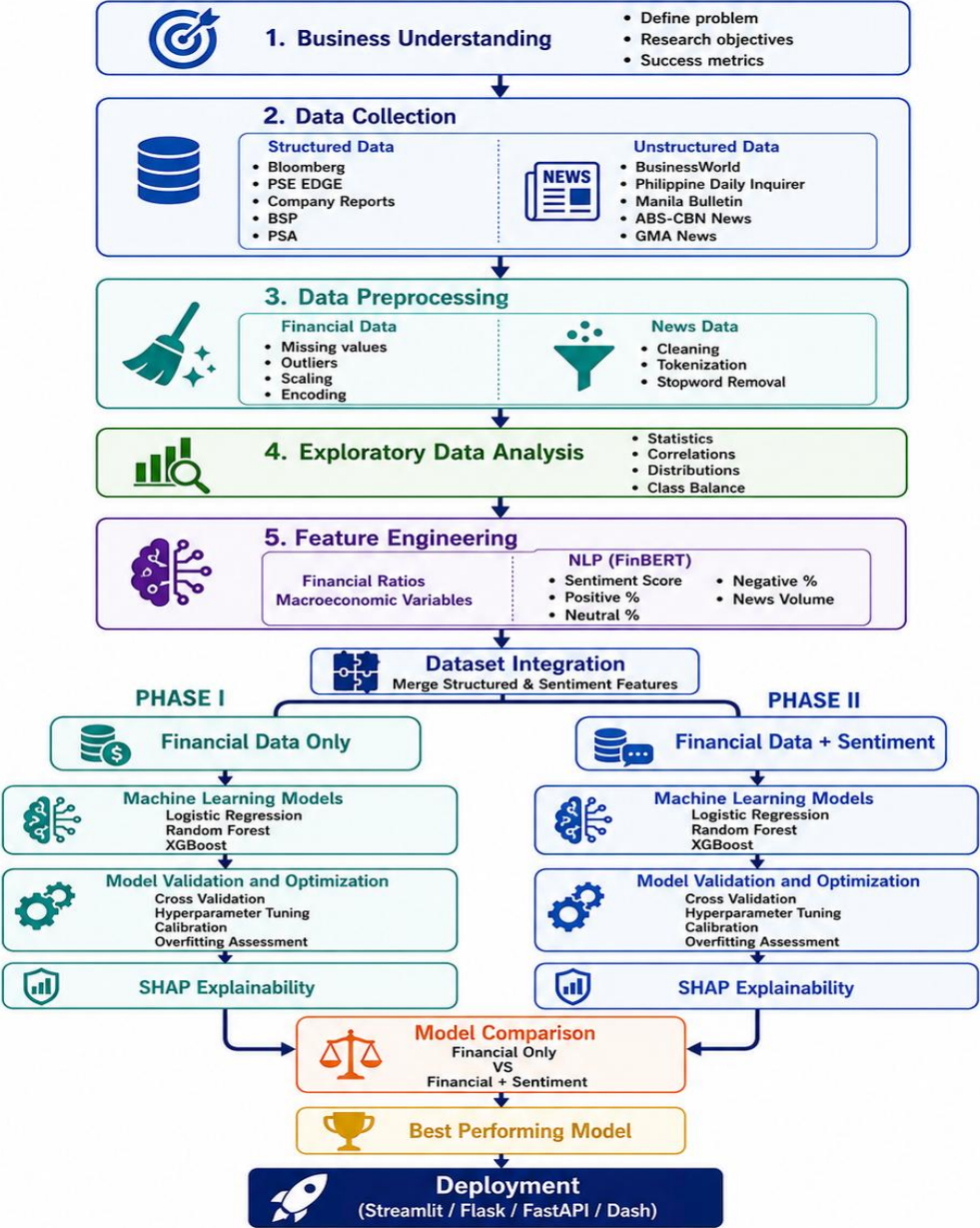
STRUCTURED PANEL DATASET: Integrated financial data from Bloomberg Terminal and official Philippine sources including PSE EDGE, PSA, and BSP.

DATASET SCOPE: Comprises 660 company-quarter observations for 30 firms spanning 2020 Q4 to 2026 Q1, totaling 38 curated variables.

ENGINEERED INDICATORS: Key metrics include liquidity, profitability, and leverage ratios, paired with macro indicators like GDP Growth, Inflation, and USD/PHP rates.

SHIFT LOGIC & LEAKAGE PREVENTION: Financial data from quarter t predicts risk for $t+1$, strictly excluding future-revealing variables (DSR_INT, TARGET_NEW, STDO).

Comparative Machine Learning Framework for Debt Servicing Risk Prediction



MACHINE LEARNING FRAMEWORK

Supervised binary classification framework to predict next-quarter (TARGET_NEXT_QTR) debt servicing risk based on historical financial indicators.

Model Selection

Logistic Regression: Interpretable baseline with L1/L2 regularization.

Random Forest: Robust ensemble against noise.

Advanced Boosting

XGBoost: High-performance gradient boosting with built-in regularization and superior scalability for financial time-series.

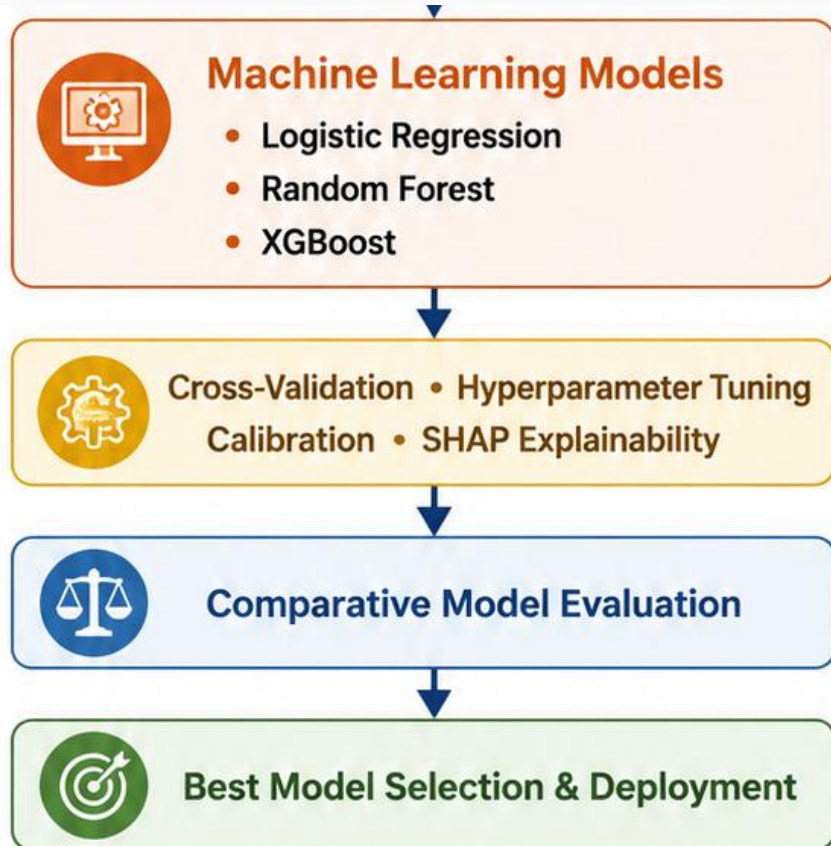
Data Strategy

Chronological Split: Training 2020 Q4–2025 Q4. Uses **TimeSeriesSplit** to prevent look-ahead bias in forecasting.

Success Metrics

Primary: ROC-AUC for discriminant ability. Supported by Brier Score, F1 Score, and Generalization Gap analysis.

PREDICTIVE MODELING & TUNING



Model Implementation:

- **Logistic Regression:** Robust baseline using L1/L2 regularization for interpretability.
- **Ensemble Methods:** Random Forest (Bagging) and XGBoost (Gradient Boosting) for non-linear pattern detection.

Preprocessing & Tuning:

- **Data Cleaning:** Median imputation for missing values and StandardScaler for feature scaling.
- **Optimization:** RandomizedSearchCV utilized for efficient hyperparameter cross-validation ROC-AUC tuning.

Performance Optimization:

- **Overfitting Control:** Implementation of Alpha and Lambda regularization in boosting models.
- **Strategic Trade-off:** Balancing baseline transparency with high-accuracy predictive robustness.

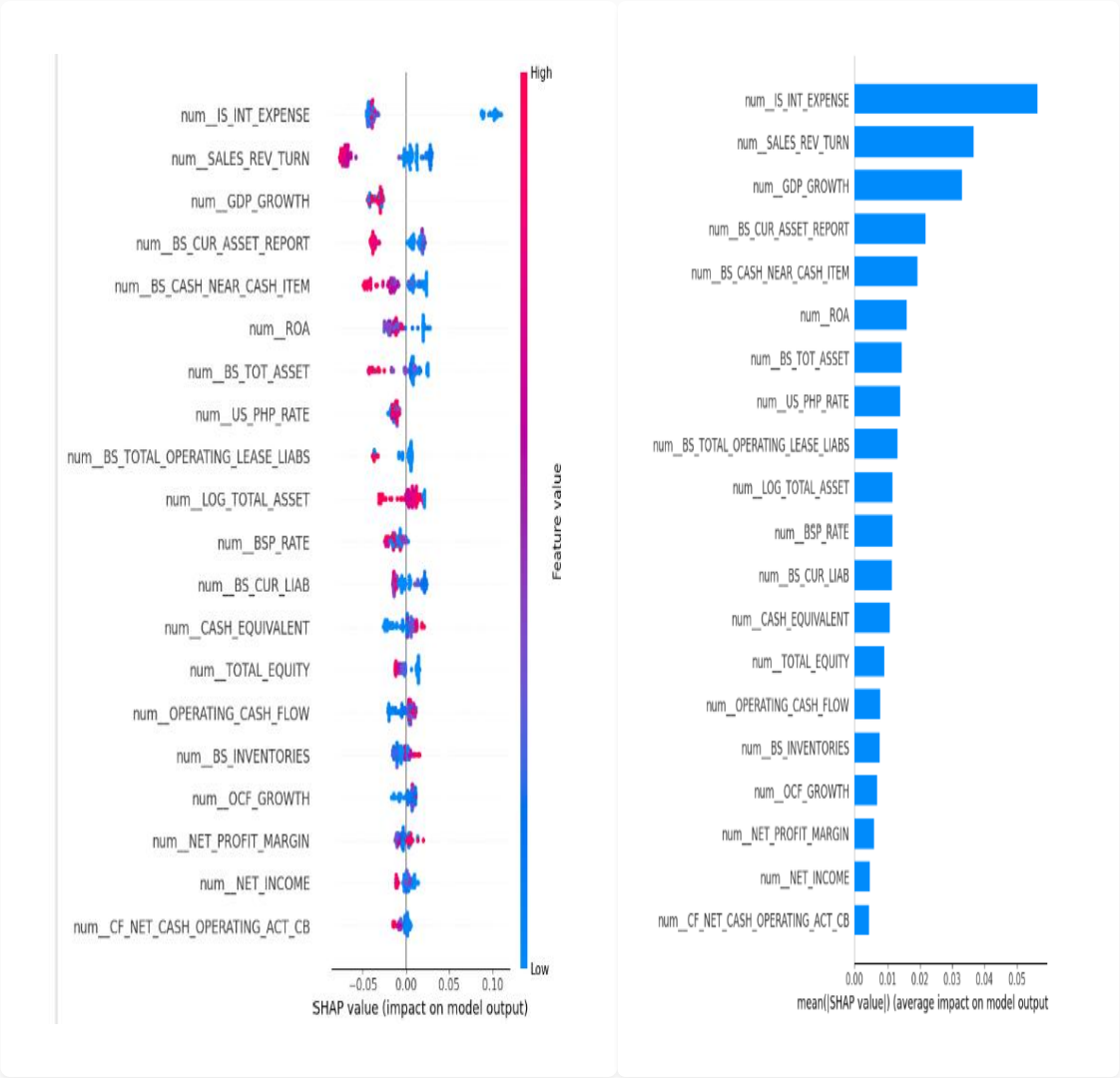
MODEL INTERPRETATION VIA EXPLAINABLE AI (XAI)

Transparency: XAI provides observation-level transparency via SHAP values, detailing feature contributions to "black-box" model predictions.

Methodology: Contrasts with global methods like MDI or Gain-based importance which offer general significance but lack individual case precision.

Business Value: Builds stakeholder trust, enhances credit evaluation, and enables proactive risk management while ensuring accountability.

Compliance: Aligns with OECD AI Principles and NIST AI Risk Management Framework regarding automated decision transparency.



STRATEGIC IMPLEMENTATION & STAKEHOLDER IMPACT



EMPOWERING STAKEHOLDERS WITH PROACTIVE INTELLIGENCE

- **Financial Institutions:** Enhance credit risk assessment and loan monitoring for high-risk borrowers.
- **Investors:** Earlier detection of vulnerability to improve portfolio risk and decision-making.
- **Corporate Managers:** Strengthen health monitoring and optimize financial planning/corrective measures.

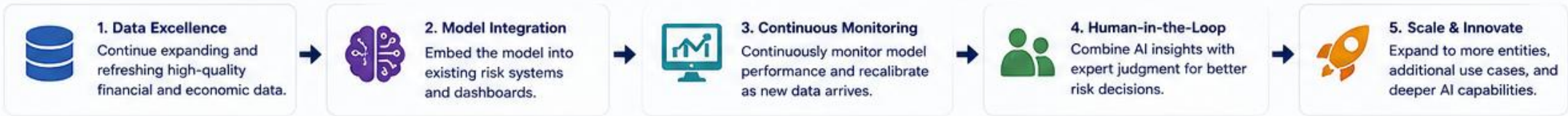
DEPLOYMENT-READY REAL-TIME MONITORING

- Deployable ML model integrated via Flask or Streamlit for continuous real-time risk monitoring.
- Leverages GDP Growth, BSP Rates, Inflation, and USD/PHP Exchange for comprehensive insights.

DRIVING EFFICIENCY & INFORMED DECISIONS

- Significant reduction in manual effort for financial analysis and credit risk assessment.
- Prioritized monitoring of high-risk entities with capability to predict 2026 Q2 debt servicing risk.

OUR STRATEGY & ROADMAP TO VALUE



End Goal

A proactive, intelligent, and scalable early warning system that drives business growth while managing risk.

